

Monotone Operators Framework

Hilbert Space $\mathcal{H}$



Linear Operator $K: \mathcal{H} \rightarrow \mathcal{H}$
Strongly Monotone



Nonlinear Operator $N_f: \mathcal{H} \rightarrow \mathcal{H}$
Monotone, Bounded



Composite: $x + KN_fx = y$



Theorem 8.8: Unique Solution

Solution x depends continuously on y